

Web Crawler - Back of the Envelope Estimation

1. Overview

Estimations help us:

- Understand **system scale**
- Design for **capacity and performance**
- Align assumptions with interviewer

2. Assumptions

- Pages crawled per month: **1 billion**
- Average page size: **500 KB**
- Data retention: **5 years**

3. Throughput Calculation (QPS)

Average QPS

None

$$\text{QPS} = 1,000,000,000 / (30 \times 24 \times 3600)$$

$$\approx 400 \text{ pages/second}$$

Peak QPS

None

$$\begin{aligned}\text{Peak QPS} &= 2 \times \text{Average QPS} \\ &\approx 800 \text{ pages/second}\end{aligned}$$

4. Storage Calculation

Monthly Storage

None

$$\begin{aligned}&1 \text{ billion pages} \times 500 \text{ KB} \\ &= 500 \text{ TB per month}\end{aligned}$$

5-Year Storage

None

$$\begin{aligned}&500 \text{ TB} \times 12 \text{ months} \times 5 \text{ years} \\ &= 30 \text{ PB (Petabytes)}\end{aligned}$$

5. Key Insights

High Throughput

- System must handle:
 - ~400 QPS (average)
 - ~800 QPS (peak)

Massive Storage

- Requires:
 - **500 TB/month**
 - **30 PB over 5 years**

Infrastructure Implications

- Distributed storage is mandatory
- Efficient compression and deduplication may be needed
- Scalable crawling architecture required

6. Key Takeaways

- Web crawling operates at **massive scale**
- Storage and throughput are **major design drivers**
- Estimations guide:
 - System architecture
 - Resource planning